

# Crew Roles & Responsibilities

Version 3.0 — Updated April 22, 2026

*"The crew is not a passenger manifest. It is a system — the most critical system aboard."*

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|-------------------------|--------------------------------------------------------------------|
| <b>Mission</b>          | Oblivion I — Crewed Mars Orbital Insertion                         |
| <b>Spacecraft</b>       | ARES-1 — 150 m x 15 m — Vertical Integrated Vehicle                |
| <b>Crew Size</b>        | 8 Astronauts (upgraded from 4 in Version 1.0)                      |
| <b>Mission Duration</b> | ~500 to 580 days in deep interplanetary space                      |
| <b>Crew Structure</b>   | 4 functional layers — 8 specialist roles — full cross-training     |
| <b>Watch Schedule</b>   | Two teams of four on alternating cycles — ARES-1 never unmonitored |
| <b>Version</b>          | Version 3.0 — supersedes Version 2.0                               |

## Section 1 — Overview

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The success of Oblivion I depends not only on the engineering of ARES-1 but on the structure, capability, and resilience of the crew that operates it. For a mission of approximately 500 to 580 days in deep interplanetary space, culminating in the first crewed Mars orbital insertion in human history, crew design is treated as a system in itself. One that must be redundant, balanced, and capable of sustaining both operational performance and human health under conditions no human being has ever faced for this duration or at this distance.

The finalized ARES-1 design — a vertical integrated vehicle 150 metres tall with a 20 metre crew capsule, 30 metre living room, and 100 metre service module — provides significantly more habitable volume than any previous deep space vehicle. This directly benefits crew psychological health across the 550-day mission. The eight crew members have access to private quarters, a dedicated medical bay, an exercise facility, and the largest common living area ever flown on a crewed spacecraft.

| Layer         | Role                                 | Position |
|---------------|--------------------------------------|----------|
| Command       | Mission Commander                    | 1 of 8   |
| Command       | Pilot / Deputy Commander             | 2 of 8   |
| Systems Eng.  | Flight Engineer (Propulsion & Power) | 3 of 8   |
| Systems Eng.  | Systems Engineer (Avionics & SW)     | 4 of 8   |
| Systems Eng.  | Mechanical & Structural Engineer     | 5 of 8   |
| Life Support  | Life Support Systems Specialist      | 6 of 8   |
| Life Support  | Flight Surgeon / Medical Officer     | 7 of 8   |
| Science & Ops | Mission Specialist (Science Ops)     | 8 of 8   |

## Section 2 — Command Layer

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The Command Layer holds ultimate authority over all mission decisions. On a 550-day deep space mission with no real-time ground support, command authority is the mechanism by which eight people with different specializations, stress responses, and national backgrounds remain a coherent operational unit.

### Mission Commander

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|  | Mission Commander                                                                                                                                                                                                                                                                                                                                                                                           |
|  | Command Layer — Position 1 of 8                                                                                                                                                                                                                                                                                                                                                                             |
|  | Ultimate authority over all mission decisions without exception. Trajectory commitment at trans-Mars injection burn, emergency response authorization, crew coordination across all four layers, communication with mission control through 20-minute delay windows, and all decisions relating to crew safety versus mission objective trade-offs. The Commander's decision is final in all circumstances. |
|  | Vehicle status monitoring across all five ARES-1 systems during watch periods. Backup navigation authority. Oversight of science program during Mars orbital operations. Public communications and mission log maintenance.                                                                                                                                                                                 |
|  | Must demonstrate proficiency in all five ARES-1 systems. Cross-trained as backup Pilot and backup Systems Engineer. Required to maintain NTP emergency shutdown procedure currency throughout mission.                                                                                                                                                                                                      |

### Pilot / Deputy Commander

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|  | Pilot / Deputy Commander                                                                                                                                                                                                                                                                                                                     |
|  | Command Layer — Position 2 of 8                                                                                                                                                                                                                                                                                                              |
|  | Vehicle maneuver execution for all planned and contingency burns. Physically executes or supervises all NTP engine ignition sequences, trajectory correction maneuvers, Mars orbital insertion burn, depot rendezvous approach burns, and trans-Earth injection burn. Maintains manual override authority for all autonomous burn sequences. |
|  | Second-in-command and backup leadership if Mission Commander becomes incapacitated. Manages crew watch schedule assignments and rotation planning. Conducts crew briefings during critical mission phases.                                                                                                                                   |
|  | Cross-trained as backup Flight Engineer with full NTP system proficiency. Required to maintain currency in all propulsion emergency procedures. Secondary training in avionics fault diagnosis.                                                                                                                                              |

## Section 3 — Systems Engineering Layer

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The Systems Engineering Layer consists of three specialists each responsible for one of the three primary technical domains that keep ARES-1 operational across 550 days. The cross-training requirement within this layer is the most demanding of any layer on the vehicle.

### Flight Engineer — Propulsion and Power

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|  | Flight Engineer (Propulsion & Power)                                                                                                                                                                                                                                                                                                                             |
|  | Systems Engineering Layer — Position 3 of 8                                                                                                                                                                                                                                                                                                                      |
|  | Full-time monitoring of the integrated NTP engine block and all power distribution systems. The NTP reactor operates continuously throughout the mission — providing thrust during burns and electrical power during cruise phases. Monitors reactor temperature, coolant flow, hydrogen fuel management, fission product buildup, and all power bus parameters. |
|  | Solar array output monitoring and optimization. Fuel cell backup system management. Li-S battery state of health tracking. NTP burn timeline preparation and engine readiness certification before every planned burn.                                                                                                                                           |
|  | Cross-trained in avionics fault diagnosis as backup Systems Engineer. Required to maintain full NTP emergency shutdown and restart procedure currency. Secondary training in structural inspection protocols.                                                                                                                                                    |

### Systems Engineer — Avionics and Software

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|  | Systems Engineer (Avionics & Software)                                                                                                                                                                                                                                                                               |
|  | Systems Engineering Layer — Position 4 of 8                                                                                                                                                                                                                                                                          |
|  | Management of all onboard computing systems, navigation, communications, and autonomous fault detection. ARES-1's triple-redundant radiation-hardened flight computers require continuous health monitoring across 550 days. Manages the Deep Space Optical Communications laser link maintaining 200 Mbps downlink. |
|  | Optical navigation system oversight — verifying position fixes daily against predicted trajectory. AI-assisted FDIR monitoring. Software patch uplink management. Autonomous Mars depot rendezvous system oversight during approach phases.                                                                          |
|  | Cross-trained as backup Flight Engineer with NTP monitoring proficiency. Required to maintain manual navigation backup procedures. Secondary training in life support sensor systems.                                                                                                                                |

### Mechanical and Structural Engineer

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|  | Mechanical & Structural Engineer            |
|  | Systems Engineering Layer — Position 5 of 8 |

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|  | Physical integrity of ARES-1 across all accessible structural elements, mechanical systems, and pressure boundaries throughout the mission. The seven-module orbital assembly creates numerous inter-module joints and seals requiring regular inspection. Micrometeorite impacts, thermal cycling, and engine burn vibrations impose cumulative structural stress. |
|  | Maintenance scheduling across all mechanical systems. Habitat module pressure integrity monitoring. Crew capsule pre-separation inspection and certification. Inter-module docking mechanism maintenance. EVA capability maintenance if external inspection required.                                                                                               |
|  | Cross-trained in power system physical maintenance as backup to Flight Engineer. Required to maintain NTP system external inspection protocols. Secondary training in life support hardware maintenance.                                                                                                                                                            |

## Section 4 — Life Support and Medical Layer

The Life Support and Medical Layer is responsible for keeping eight human beings alive and functional across 550 days. The finalized ARES-1 design provides dedicated spaces for each life support function — the 30 metre living room section houses the crew quarters, medical bay, and exercise facility, while the service module houses the ECLSS machinery in a dedicated bay separated from living areas.

### Life Support Systems Specialist

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|  | Life Support Systems Specialist                                                                                                                                                                                                                                                                                                                                                                             |
|  | Life Support & Medical Layer — Position 6 of 8                                                                                                                                                                                                                                                                                                                                                              |
|  | Management of the complete closed-loop ECLSS serving eight crew members — the highest biological load ever placed on a deep space life support system. Sabatier CO <sub>2</sub> recycling reactor, solid oxide oxygen electrolysis, vapor compression distillation water recovery achieving greater than 98% efficiency, atmospheric management, thermal comfort control, and aeroponic food growing units. |
|  | Consumables inventory management tracking consumption rates against mission timeline projections across 550 days. Waste processing system management. Crew hygiene facility maintenance. Radiation storm shelter integrity verification.                                                                                                                                                                    |
|  | Cross-trained in mechanical systems maintenance as backup to the Mechanical Engineer for life support hardware interventions. Required to maintain emergency atmospheric management procedures for rapid response to pressure loss.                                                                                                                                                                         |

### Flight Surgeon / Medical Officer

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|  | Flight Surgeon / Medical Officer                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|  | Life Support & Medical Layer — Position 7 of 8                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  | Complete responsibility for physical and psychological health of all eight crew members across the full mission duration. Daily assessments covering bone density, muscle mass, cardiovascular fitness, radiation dose tracking, nutrition, and sleep quality. Psychological assessments covering crew morale, interpersonal dynamics, stress indicators, and early identification of psychological degradation. Must be capable of emergency surgery in microgravity. |
|  | Crew exercise program design, implementation, and enforcement — the mandatory two-hour daily exercise regime is medical prescription. Pharmacological inventory management. Telemedicine consultation coordination with Earth-based medical teams. Psychological intervention during return transit when crew stress peaks.                                                                                                                                            |
|  | Cross-trained as backup Life Support Systems Specialist with full ECLSS monitoring proficiency. Required to maintain emergency life support override procedures.                                                                                                                                                                                                                                                                                                       |

## Section 5 — Science and Mission Operations Layer

The Science and Mission Operations Layer is responsible for the scientific objectives that justify the entire program investment. The Mission Specialist carries the scientific legacy of Oblivion I — the data returned from Low Mars Orbit will exceed anything any robotic orbiter has ever produced.

### Mission Specialist — Scientific Operations

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|  | Mission Specialist (Scientific Operations)                                                                                                                                                                                                                                                                                                                                                                                      |
|  | Science & Mission Operations Layer — Position 8 of 8                                                                                                                                                                                                                                                                                                                                                                            |
|  | Direction of the complete scientific program across all mission phases. Outbound transit: deep space science instruments, cosmic ray flux measurements, solar wind characterisation, interplanetary dust sampling. Mars orbital operations: high-resolution multispectral surface imaging, atmospheric probe deployment, magnetosphere characterisation, deployment of fourteen CubeSat orbiters in two precisely timed salvos. |
|  | Science instrument calibration, maintenance, and operation throughout the mission. Data management and compression for Deep Space Optical Communications transmission. Science timeline coordination with coalition science teams on Earth. Return transit science operations.                                                                                                                                                  |
|  | Cross-trained as backup Systems Engineer for avionics monitoring. Required to maintain communications system operational procedures as backup to the Systems Engineer for DSOC link management.                                                                                                                                                                                                                                 |

## Section 6 — Redundancy and Cross-Training

| Primary Role             | Primary Function                   | Cross-Trained Backup For                          |
|--------------------------|------------------------------------|---------------------------------------------------|
| Mission Commander        | Command & mission authority        | Pilot / Deputy Commander, Systems Engineer        |
| Pilot / Deputy Commander | Maneuver execution, backup command | Flight Engineer (NTP), Avionics backup            |
| Flight Engineer          | NTP system & power                 | Systems Engineer (avionics), Mechanical Engineer  |
| Systems Engineer         | Avionics, nav & software           | Flight Engineer (power), Life Support sensors     |
| Mechanical Engineer      | Structural integrity & repairs     | Flight Engineer (hardware), LS hardware           |
| Life Support Specialist  | ECLSS management                   | Mechanical Engineer (LS hardware), Medical backup |
| Flight Surgeon           | Crew health & psychology           | Life Support Specialist (ECLSS monitoring)        |

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|--------------------|--------------------|-------------------------------------------|
| Mission Specialist | Science operations | Systems Engineer (avionics), Comms backup |
|--------------------|--------------------|-------------------------------------------|

### Watch Schedule — Team Alpha and Team Beta

| Team       | Crew Members                                                                    | Coverage                                                                                                                    |
|------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Team Alpha | Commander, Flight Engineer, Life Support Specialist, Mission Specialist         | Primary watch outbound transit odd-numbered days. Both teams active during all critical burns and Mars orbital operations.  |
| Team Beta  | Pilot / Deputy Commander, Systems Engineer, Mechanical Engineer, Flight Surgeon | Primary watch outbound transit even-numbered days. Both teams active during all critical burns and Mars orbital operations. |

## Section 7 — Crew Selection Philosophy

The Oblivion I crew has not yet been selected. The Mission Commander will almost certainly be American given the US lead role. The remaining seven seats will be distributed among coalition partners through a combination of technical qualification and diplomatic negotiation. No coalition partner contributing at the \$50 billion level or above can reasonably be excluded if a qualified candidate exists.

Every crew member who boards Oblivion I must understand, with full clarity, what the mission's failure clause means. The program they are joining has no safety net. If the mission fails, eight nations absorb a trillion dollars in unrecoverable debt and the era of international deep space cooperation ends before it truly began. The crew carries that weight for 550 days. Selection must verify all eight candidates are capable of carrying it.

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**Their structure and training are engineered to ensure that human limitations never become the weakest link in the mission architecture.**

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*End of Document — Crew Roles & Responsibilities — Version 3.0 Next revision triggered by: Crew Selection Completion*